

Week 7 — Missing Data: What Multiple Imputation Can and Cannot Fix

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

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Tuesday lecture — precedes Thursday lab L6 and increment P5

Learning objectives

- ▶ Classify each reduction by **type** first — design, eligibility, target population, outcome ascertainment, missingness — then apply MCAR / MAR / MNAR to the missingness only
- ▶ Tell **design** apart from **missingness** — the two losses students most often conflate: selection into the fasting subsample is **design**, answered by the weight and a target-population sentence; item-missing confounders inside the cohort are the **missingness**, and only that is a missing-data problem
- ▶ Contrast **complete-case**, a **model-free extreme-value stress test**, and **multiple imputation** on the **replication contrast (IV vs I)**, **design-aware Model 1: 2.97 → 2.98** — on the **locked scientific contrast (IV vs III)** the same sensitivity analysis never leaves **1.29**
- ▶ Specify a defensible imputation model: variables in, number of imputations M , and **Rubin's rules** for pooling
- ▶ Judge what MI **can** repair (MAR item-missingness) and **cannot** (structurally unobserved exposure)

Continuity — from weighting to missingness

- ▶ **Last week (Wk6, L5):** design-aware estimate — NHANES **weights + strata + PSU** pulled Group IV from naive ~**15** toward ~**12.4**
- ▶ Weighting fixed *representation*; it did **not** fill in *missing values*
- ▶ **Today:** confront values that are simply absent — separate what MI can help vs. what it cannot
- ▶ Last interrogation method before M1 compiles all five — the shift from “*find the flaws*” to critique is M2 (Wk 9), and improvement is defended at M3 (Wk 11)
- ▶ Sets up **L6** (pattern → MI → design-adjusted refit) and **P5** (your paper’s missing-data sensitivity analysis)

Where we are in the project

- ▶ The arc: **Replicate + Interrogate** (survey design → missing data) → **M1** compiled and presented **next week** → **Critique** another group's M1 (**M2**, Wk 9) → **Improve & defend** (**M3**, Wk 11)
- ▶ Missing data is the **fifth and final threat lens** before students write their critique of another group's M1
- ▶ Today's method interacts with everything prior: **impute** → **weight** → **refit**
- ▶ Hold two cohort sizes in mind: the **full analytic file** is **6,371** and the **locked domain** is **6,048** — the paper reports about **6,300**
***MASLD:** the reproduction lands on 6,371, not the paper's approximate 6,300 — so where did the other adults go?*

Why missingness is bias, not bookkeeping

- ▶ Dropping incomplete rows silently changes *who* is in your analysis — the **estimand shifts** even if nothing is mis-measured
- ▶ The *amount* missing bounds **power**; the *mechanism* determines **bias**
- ▶ **5% MNAR** can mislead more than **30% MCAR**
- ▶ Punchline preview: one loss MI handles well (item nonresponse), one it cannot touch because it is **design**, not missing data
***MASLD:** the paper counts **585** all-cause deaths, our reproduction **586**; those deaths drive the survival estimates — if the missing are systematically higher- or lower-risk, every HR moves.*

The taxonomy: MCAR / MAR / MNAR

- ▶ **MCAR** — missingness independent of everything; complete cases are a random subsample: unbiased but wasteful
- ▶ **MAR** — missingness explained by *observed* data; recoverable by conditioning / imputing on those observables
- ▶ **MNAR** — missingness depends on the *unobserved value itself*; not fixable from data alone
- ▶ These labels are **untestable** from the data — argue the mechanism substantively, then run sensitivity analysis
MASLD: which label fits the **792 adults inside the fast-ing subsample who still lack an FLI**? Those never sampled into the subsample are a different question entirely.

A picture of the taxonomy

- ▶ Three arrows into the “missing” indicator: from **nothing** (MCAR) / from **observed X** (MAR) / from **the value itself** (MNAR)
- ▶ The one diagnostic you *can* do: compare observed characteristics of **complete vs. incomplete** cases
- ▶ That comparison can make MCAR *less* plausible — but **cannot** confirm MAR vs. MNAR
- ▶ Complete-case analysis is unbiased only under **MCAR** (or MAR in covariates, not outcome-dependent)
***MASLD:** the complete-case vs. imputed contrast we run later is this diagnostic, live.*

Classify first: which reductions are actually missing data?

- ▶ **FLI** needs triglycerides + GGT + BMI + waist — and **triglycerides** come only from the NHANES **fasting subsample** (GGT and anthropometry are MEC-wide); one fasting input binds the whole index
- ▶ **After eligibility**, the two largest reductions are **designed subsampling** and **disease definition** — neither is missing data
- ▶ Genuine missingness across the whole funnel: **792 item-nonresponse losses**. A further **2 records fail outcome ascertainment** — a separate category, not missingness, and not added to them. Inside the **locked domain**, Model-1 missingness is **1.8%** — all of it cancer status
- ▶ Upstream of any confounder adjustment — a scope question, not a missingness one
***MASLD:** this is why N collapses from all NHANES adults to the **full analytic file** (6,371) and the **locked domain** (6,048) — the phenotype simply cannot be assigned to most adults.*

Why designed subsampling is not MNAR

- ▶ **Fasting status is not random** — it tracks appointment time, diabetes, medication, age: variables tied to metabolic risk
- ▶ NHANES **designed** the fasting subsample and supplies WTSAF2YR for it — absence there is **not, by itself**, evidence of MNAR
- ▶ The estimate then speaks to the **fasting subpopulation**, not all adults
- ▶ MI cannot invent a measurement never taken — but that argues for the **right weight**, not for declaring the study irreparable
- ▶ Honest fix: **scope language** (“among adults with ascertainable MASLD”) + a selection / sensitivity argument
MASLD: *back to M1’s circularity — the fasting filter and the phenotype **share BMI + waist inputs**, so selection and exposure definition are entangled.*

Break

10-minute break

On return: we move from the loss **MI cannot repair** — design, which the **weight** answers — to the one it **can**.

Item nonresponse inside the cohort

- ▶ Within the **locked domain** — the **6,048** rows P5 analyses — confounders are only partly complete
- ▶ **Income ~9% missing**, blood pressure **~3%** — but *neither is in the paper's Model 1*
- ▶ Inside **Model 1**, exactly one covariate is incomplete: **cancer status, 109 rows, 1.8%**
- ▶ Classic **item-missing covariates** — the setting MI was designed for, plausibly **MAR** given NHANES's rich covariates
- ▶ Contrast with the design and disease steps: here the **rows exist**, exposure and outcome are **observed** — only some predictors are blank
***MASLD:** on the paper's Model 1, complete-case quietly drops **1.8%** of the cohort — 109 people, for a blank cancer-status field, not anything about liver fat. (Adding income and BP would make it ~12.8% — but that changes the adjustment set, and answers a different question.)*

Complete-case analysis: the hidden default

- ▶ Most regression software **listwise-deletes** — the analyst rarely chooses it consciously
- ▶ Costs: **lost precision**, plus **bias** whenever covariate missingness is associated with the outcome
- ▶ Defensible only under **MCAR** — or **MAR in covariates, not outcome-dependent** — assumptions you usually cannot justify
- ▶ The reported estimate reflects **whoever survived deletion** — an unstated subsample
***MASLD:** the paper's adjusted Group IV HR reflects whoever survived listwise deletion — a subsample it never names.*

Multiple imputation: the idea in three steps

- ▶ **Impute** — draw M completed datasets from a model predicting each missing value from the observed ones (including the outcome)

MASLD: the replication contrast (IV vs I), design-aware Model 1 — complete-case **2.97**, both extreme assignments **2.98 / 2.99**, MI **2.98**. Everything inside a 0.02 band.

Multiple imputation: the idea in three steps

- ▶ **Impute** — draw M completed datasets from a model predicting each missing value from the observed ones (including the outcome)
- ▶ **Analyze** — fit the *same* substantive model in each of the M datasets

MASLD: the *replication contrast (IV vs I)*, *design-aware Model 1* — complete-case **2.97**, both extreme assignments **2.98 / 2.99**, MI **2.98**. Everything inside a 0.02 band.

Multiple imputation: the idea in three steps

- ▶ **Impute** — draw M completed datasets from a model predicting each missing value from the observed ones (including the outcome)
- ▶ **Analyze** — fit the *same* substantive model in each of the M datasets
- ▶ **Pool** — combine with **Rubin's rules**: within- + between-imputation variance carries the uncertainty of not knowing the true value
***MASLD: the replication contrast (IV vs I), design-aware Model 1** — complete-case 2.97, both extreme assignments 2.98 / 2.99, MI 2.98. Everything inside a 0.02 band.*

Reading a sensitivity analysis that changes nothing

- ▶ Complete cases were a slightly **higher-risk slice**; restoring the dropped **1.8%** removes that selection **if MAR holds** — there is no observed truth here to check it against
- ▶ What MI actually did: estimate **2.97** → **2.98**, and the interval came back marginally **wider**, not narrower — the imputed values have almost no leverage
- ▶ Caution: MI moves the estimate only as much as the **missing predictors matter** — *a null move is also a finding*
- ▶ **The wording that matters:** *this missing-data problem mattered little **after we measured it**. The method was necessary because we did not know that in advance*
MASLD: *“missing data don’t matter” is the wrong lesson. The right one: measure before you presume, and report a null sensitivity analysis as the result it is.*

Building a defensible imputation model

- ▶ **Include the outcome** — for survival, the Nelson–Aalen cumulative hazard + event indicator; omitting it biases associations toward null
- ▶ Be **inclusive**: all analysis-model covariates + useful auxiliary predictors of missingness (**congeniality**)
- ▶ **Match variable types** — logistic / **PMM** for skewed or bounded variables; respect the survey design where feasible
- ▶ A stripped-down imputation model is a *silent* source of bias
MASLD: *income and BP should be imputed from age, sex, BMI, waist, comorbidity* **AND the mortality outcome** — *not from a stripped-down model.*

How many imputations, and how to report

- ▶ Rule of thumb: **M** the % of incomplete cases → roughly **10–20** here; more is cheap insurance, too few is false precision
- ▶ Report: **mechanism assumed, variables imputed, M, software/method**, and the pooled estimate with **Rubin SEs**
- ▶ Always show **complete-case vs. MI side by side** — divergence is a sensitivity signal, agreement is reassurance
MASLD: a small table on the replication contrast (IV vs I), design-aware Model 1 (CC 2.97 / stress 2.98–2.99 / MI 2.98) is the P5 deliverable in miniature — and the **survey design** moved this estimate far more than the missingness did.

What MI fixes vs. what it cannot

- ▶ **Fixes (item nonresponse):** MAR item-missing confounders inside the cohort — recovers N and, **under MAR with a correctly specified imputation model**, removes complete-case selection bias
- ▶ **Not a missing-data problem at all:** those never sampled into the fasting subsample — that is scope, addressed by the correct weight and an honest target-population statement
- ▶ Designed subsampling is fixed by the **correct weight** and a clear target-population statement, not by imputation
- ▶ Corollary: **MNAR item-missingness** also needs sensitivity analysis (e.g., **delta-adjustment**), not naive MI
- ▶ Back to the gate: the categories MI cannot reach are the categories **M0 criterion 6** does not accept on their own **MASLD**: *the external-validity ceiling is a **scope** statement about the fasting subsample — the correct weight addresses it; imputation was never the tool.*

Putting the interrogation together

- ▶ Sequence for a design- and missingness-aware refit: **impute (item nonresponse) → weight / strata / PSU (Wk6) → fit → pool**
- ▶ Each threat moves Group IV a *defensible* direction — report the **stack**, not one number
- ▶ Four headline numbers — **15.1 naive**, **~2.9 adjusted**, **~12.4 weighted-crude**, **2.98 imputed** — that are *not* a sequence: different models, different row sets
- ▶ To separate **adjustment** from **design**, hold the rows fixed — P4's 2×2 on the same 5,939 rows: weighting at Model 1 moves **3.79 → 2.97**
- ▶ Next week (M1): students compile the five lenses and name the dominant threat, not which fixes *their* paper needs — today supplies the missing-data piece
MASLD: discuss why **order and combination** matter — *the number you report is a stack of defensible choices, not a single answer.*

Bridge to today's lab (L6) and P5

- ▶ In **L6** (EpiMethods missing-data module) students will:
 - ▶ (1) tabulate the missingness pattern for the fixed dataset and name the likely mechanism per variable
 - ▶ (2) run **MI** with a properly specified model and a justified M
 - ▶ (3) refit the survival model with **Rubin pooling**, re-apply the Wk6 survey design, report **CC vs. MI vs. design-adjusted** side by side
- ▶ Practice telling **design** apart from **missingness**: which gaps MI addresses vs. which are scope limits to write up
- ▶ **P5 (your locked paper)**: a missing-data sensitivity analysis — document each variable's mechanism, run MI where MAR is defensible, contrast with complete-case, and state plainly what imputation **cannot** recover
***MASLD**: the paper reports one HR from an unstated sub-sample — P5 asks you to show your number under missingness stress and name its true validity ceiling.*